

SFC-008 — LLM Scoring as Implicit SFC

Statistical Ranking, Patchwork Support, and the Limits of Implicit Feasibility Confidence

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Repository: <https://github.com/sizhet/Structural-Feasibility-Confidence-SFC>

1. Introduction

Structural Feasibility Confidence (SFC) was introduced as a framework for explaining how an intelligent system can form disciplined confidence about a target that has not been fully executed before in exactly the same form.

At first, SFC may appear most naturally connected to human and engineering examples:

- a craftsman quoting a custom commission,
- an engineer estimating an unseen implementation,
- a Task Calling Graph being proposed before all Action Calling Graphs exist,
- or an AI runtime deciding whether a gap is bridgeable.

However, there is another important connection:

LLM and Transformer scoring already contain a weak, implicit form of SFC-like logic.

When a language model scores, ranks, or selects candidate continuations, it is not performing explicit Structural Feasibility Confidence in the full sense.

It does not usually maintain explicit confirmed zones, extension zones, patchwork proof structures, gap types, or commitment policies.

But it does perform something structurally related:

It evaluates candidate continuations according to whether they are statistically and contextually supported by learned patterns, local coherence,

semantic adjacency, and trajectory compatibility.

In this sense, LLM scoring can be interpreted as an **implicit statistical precursor** to SFC.

The purpose of this document is to clarify that relationship.

2. LLM Scoring Is Not Explicit Proof

A language model does not usually know that a candidate answer is feasible in the same way an engineer may know that a system architecture is feasible, or a craftsman may know that a custom object is buildable.

It usually lacks:

- explicit execution proof,
- explicit material constraints,
- explicit task-action grounding,
- explicit failure-mode analysis,
- explicit commitment boundary control.

Therefore, it would be a mistake to say that LLM scoring is already full SFC. A high-probability continuation is not the same thing as a verified feasible plan.

A fluent answer is not the same thing as a supported commitment.

A coherent pattern is not the same thing as a bridgeable gap.

This distinction is essential.

However, rejecting the equivalence does not mean ignoring the structural analogy.

3. The Core Analogy

When an LLM ranks candidate continuations, it implicitly asks something like:

- Does this token fit the prior tokens?
- Does this phrase match known discourse patterns?
- Does this answer form resemble successful examples in training?
- Does the continuation remain semantically coherent?
- Does it stay inside the current context trajectory?

- Does it align with latent task format, genre, and expected completion style?

These questions are not explicit SFC questions.

But they resemble a statistical version of structural support evaluation.

A candidate continuation is favored when it lies inside a dense region of learned support.

It is disfavored when it appears unsupported, incoherent, or too far from the active context trajectory.

This is why LLM scoring can be interpreted as a primitive form of:

Implicit Structural Feasibility Confidence

It is implicit because the support is not exposed as a structured proof.

It is statistical because the support is learned from distributional patterns.

It is feasibility-like because it ranks what appears continuable, coherent, and locally supportable.

4. LLM Scoring as Statistical Patchwork Support

The analogy becomes clearer when viewed through Patchwork Feasibility Proof.

A model's preferred continuation is rarely supported by one exact precedent. Instead, it is supported by many partial signals:

- token-level compatibility,
- phrase-level precedent,
- syntactic continuity,
- semantic adjacency,
- discourse-level coherence,
- task-format similarity,
- instruction-following patterns,
- latent structural analogy,
- and prior examples distributed across training.

This resembles a statistical version of Patchwork Feasibility Proof.

The model does not say:

"This exact continuation has already been proven."

Rather, it behaves as if saying:

"This continuation is supported by many overlapping partial precedents and local compatibility signals."

That is very close to the spirit of PFP:

- no single full precedent,
- many partial supports,
- assembled confidence,
- local-to-global plausibility.

The difference is that in LLM scoring, the patchwork remains mostly latent, untyped, and ungoverned.

5. Token Probability as a Weak Support Field

A language model's probability distribution can be interpreted as a weak support field over possible continuations.

High-probability candidates often correspond to continuations with strong statistical support from:

- context,
- learned patterns,
- common task formats,
- local syntax,
- semantic coherence,
- and prior discourse trajectories.

Low-probability candidates often correspond to continuations with weaker or less coherent support.

This resembles the idea of Function Tunnel Confidence Thickness:

- dense support near familiar continuation paths,

- thinner support near unusual but still possible extensions,
- weak support in remote or incoherent regions.

But again, the analogy has limits.

LLM probabilities are not direct feasibility measures.

They are continuation likelihoods under a learned model.

They may correlate with feasibility in many textual or pattern-rich domains, but they do not guarantee truth, action success, or external-world validity.

This is why SFC must go beyond probability.

6. What LLM Scoring Gets Right

LLM scoring already captures several ingredients that are important for SFC.

6.1 Local Coherence

The model evaluates whether a continuation fits the immediate context.

6.2 Pattern Adjacency

The model can extend from known patterns to nearby unseen patterns.

6.3 Distributed Precedent

The model can use many partial precedents rather than one exact example.

6.4 Format Sensitivity

The model can recognize task formats, rhetorical structures, code patterns, and document conventions.

6.5 Soft Ranking

The model naturally ranks alternatives rather than making only binary decisions.

These capabilities are important because SFC also requires graded support judgment rather than exact-precedent-only logic.

7. What LLM Scoring Lacks

Despite the analogy, LLM scoring lacks several features required for explicit SFC.

7.1 Explicit Zone Classification

LLMs do not naturally label a candidate as belonging to:

- Confirmed Zone,
- Extension Zone,
- or Speculative Zone.

7.2 Explicit Gap Typing

They do not reliably identify whether a weakness is caused by:

- missing evidence,
- missing mechanism,
- missing action path,
- missing tool,
- missing causal model,
- or missing task decomposition.

7.3 Explicit Bridgeability Analysis

They may generate a plausible bridge without knowing whether the bridge is actually executable, testable, or recoverable.

7.4 Explicit Commitment Governance

They may produce confident language even when structural support is thin.

7.5 External-World Grounding

They may score a continuation as likely even when the claim is false, outdated, physically impossible, or operationally infeasible.

These limitations explain why implicit SFC is not enough.

8. From Implicit Scoring to Explicit SFC

The SFC framework can be understood as an attempt to make the implicit support logic of generative models more explicit, inspectable, and controllable.

Instead of merely asking:

"Which continuation has the highest score?"

an SFC-aware runtime asks:

- What support does this candidate have?
- Is the support confirmed or merely extension-based?
- Which parts are backed by direct precedent?
- Which parts rely on patchwork support?
- What gaps remain?

- Are those gaps bridgeable?
- What commitment policy is justified?

This is a major shift.

LLM scoring provides ranked candidates.

SFC adds structural accountability.

9. LLMs as Generators, SFC as Governance

A useful design principle is:

LLMs can generate candidate continuations, plans, decompositions, and bridges.

SFC can evaluate how structurally supported those candidates are.

In this view, an LLM is not rejected.

It is repositioned.

The LLM becomes a powerful generator of possibilities.

SFC becomes a governance layer that asks:

- Which generated candidates lie inside supported structure?
- Which candidates are plausible but unsupported?
- Which require retrieval, tool use, testing, or human review?
- Which should be downgraded from commitment to proposal?
- Which should be rejected as speculative?

This division of labor is important for future AI systems.

It preserves the creativity and pattern fluency of LLMs while adding a structural discipline layer.

10. Implicit SFC and Hallucination

The SFC view also helps explain hallucination.

A hallucinated answer may be locally supported by language patterns but globally unsupported by reality, evidence, or executable structure.

In SFC terms, hallucination often occurs when:

- statistical continuation support is mistaken for feasibility support,
- local patchwork signals are mistaken for global proof,

- extension confidence is treated as confirmed confidence,
- speculative completion is expressed as established fact.

This is not merely a content error.

It is a failure of commitment governance.

The model has generated something that appears structurally fluent, but the system has not checked whether the support is strong enough for the level of commitment expressed.

SFC can help by separating:

- candidate generation,
- support evaluation,
- gap typing,
- and commitment expression.

11. Implicit SFC and Ranking-Based Intelligence

The connection to LLM scoring also suggests a broader point:

Many intelligent systems may begin with ranking before they acquire explicit proof.

A brain, a craftsman, or a model may first learn to sense:

- this path feels supported,
- that move feels risky,
- this completion fits,
- that continuation breaks pattern,
- this target lies near known skill,
- that target is outside current reach.

Only later may these judgments become explicit, inspectable, and formalized.

In that sense, SFC may be seen as a structural upgrade of an older and more

primitive ranking capability:
from implicit support ranking
to explicit feasibility confidence.

This is an important bridge between statistical AI and structural intelligence.

12. Implications for Future AI Runtime Design

An SFC-aware AI runtime may combine LLM scoring with explicit support analysis.

A possible architecture:

1. **Generate candidates** using LLMs or other generative models.
2. **Extract structure** from the candidates:
 - task nodes,
 - dependencies,
 - claims,
 - required tools,
 - implied action paths.
3. **Locate support:**
 - confirmed precedents,
 - reusable modules,
 - retrieved evidence,
 - neighboring trajectories.
4. **Identify gaps:**
 - missing data,
 - missing proof,
 - missing action operators,
 - missing causal mechanisms.

5. **Classify confidence zone:**

- confirmed,
- extension,
- speculative.

6. **Select commitment policy:**

- answer,
- qualify,
- probe,
- call tools,
- stage,
- refuse.

This turns LLM scoring from a final authority into an input to structured runtime governance.

13. **Why This Matters for SFC**

This document strengthens the SFC program in two ways.

First, it shows that SFC is not alien to current AI systems.

Modern LLMs already perform a weak statistical version of support-sensitive ranking.

Second, it shows why SFC is still needed.

Implicit scoring is powerful but insufficient.

It needs to be made explicit, typed, inspectable, and tied to commitment policy.

This relationship is important because it prevents two simplistic views:

- "LLMs already solve this because they score candidates."
- "LLMs have nothing to do with SFC because they lack explicit proof."

The better view is:

LLM scoring is an implicit, statistical, ungoverned precursor to SFC.

SFC is a structural, explicit, governance-oriented extension of that capability.

14. Conclusion

LLM and Transformer scoring can be interpreted as a primitive form of implicit Structural Feasibility Confidence.

When a model ranks candidate continuations, it evaluates whether they are supported by context, prior patterns, semantic adjacency, discourse coherence, and latent structural similarity.

This resembles a statistical version of Patchwork Feasibility Proof.

But implicit scoring is not enough.

It lacks explicit zone classification, gap typing, bridgeability analysis, external-world grounding, and commitment governance.

Therefore, SFC should not be seen as redundant with LLM scoring.

It should be seen as a structural upgrade:

from likely continuation

to supported commitment.

That upgrade is essential if future AI agents are to move from fluent generation toward disciplined runtime intelligence.

Key Takeaways

- LLM/Transformer scoring can be interpreted as a weak, implicit form of SFC-like support ranking.
- Candidate continuations are often supported by many partial signals, resembling statistical Patchwork Feasibility Proof.
- LLM scoring is useful but not sufficient because it lacks:
 - explicit zone classification,
 - explicit gap typing,
 - bridgeability analysis,
 - external-world grounding,
 - and commitment governance.
- SFC can turn implicit ranking into explicit runtime judgment.
- A useful design principle is:
 - **LLM generates candidates; SFC governs commitment.**

